

PREPAURA

Placement & Training Platform

Computer Science & Engineering

4th Year Technical MCQ Test - KTU 2019-scheme aligned practice bank

Questions	10	Total Marks	10
Duration	20 minutes	Passing Marks	5
Difficulty	Hard	Question Type	MCQ

Candidate Name		Registration No.	
Date		Score	

Instructions

1. Choose the single best answer for each question.
2. Each question carries 1 mark. There is no negative marking in this practice bank.
3. The answer key is intentionally omitted from this resource copy.

Question Paper

1. In alpha-beta pruning, when a MAX node obtains a value equal to or greater than the current beta value inherited from its ancestor, the remaining children can be pruned because:

- A. MAX is minimizing
- B. The minimizing ancestor will never choose this branch
- C. The heuristic becomes admissible
- D. The tree becomes acyclic

2. In a distributed system, Lamport logical clocks guarantee that if event a causally precedes event b, then:

- A. $C(a) = C(b)$
- B. $C(a) > C(b)$
- C. $C(a) < C(b)$
- D. The timestamps are always consecutive

3. A distributed mutual-exclusion algorithm that requires permission from every other process is vulnerable to a single process failure primarily because:

- A. It has no timestamps
- B. The failed process can block permission acquisition
- C. Messages are always duplicated
- D. Critical sections become nested

4. For RSA, if p and q are distinct primes and $n = pq$, the private exponent d is chosen so that:

- A. $d = e \bmod n$
- B. $ed \equiv 1 \bmod \phi(n)$
- C. $d = \phi(n) + e$
- D. $ed \equiv 0 \bmod \phi(n)$

5. For a cryptographic hash function used for collision-resistant signatures, an attacker finding $m \neq m'$ with $H(m)=H(m')$ would directly violate:

- A. Confidentiality only
- B. Collision resistance
- C. Availability
- D. Forward secrecy

6. In a data-mining decision tree, choosing the attribute with maximum information gain tends to:

- A. Increase expected entropy after splitting
- B. Reduce impurity of the child nodes
- C. Guarantee zero test error
- D. Guarantee a balanced tree

7. DBSCAN can identify a point as a core point when it has at least minPts points within:

- A. The entire database
- B. Its epsilon-neighborhood
- C. The convex hull
- D. A random sample

8. In MapReduce, a mapper emits intermediate key-value pairs and the shuffle phase primarily:

- A. Executes SQL joins directly
- B. Partitions and groups intermediate values by key for reducers
- C. Encrypts all input data
- D. Deletes duplicate files

9. An SVM with a Gaussian RBF kernel can model nonlinear decision boundaries because the kernel:

- A. Explicitly computes every feature of infinite dimension
- B. Computes an inner product in an implicit feature space
- C. Removes the margin constraint
- D. Forces all points onto one line

10. In cloud computing, the main distinction of Infrastructure as a Service is that the customer typically manages:

- A. Only the physical data-center cooling
- B. The guest OS and applications while the provider manages physical infrastructure
- C. Only the provider's hypervisor source code
- D. Only DNS root servers

End of Test